RAG Learning

1.3 Coordinate Geometry

Straight Lines

Key Formulae

Formula	Expression
Gradient	$m = \frac{y_2 - y_1}{x_2 - x_1}$
Slope-intercept form	$y = mx + c$
General form	$ax + by + c = 0$
Point-gradient form	$y - y_1 = m(x - x_1)$
Midpoint	$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Example: Find the equation of the line passing through $(2, 5)$ and $(6, 13)$.

Gradient: $m = \frac{13 - 5}{6 - 2} = \frac{8}{4} = 2$

Using point-gradient form with $(2, 5)$:

$y - 5 = 2(x - 2) \Rightarrow y = 2x + 1$

Parallel and Perpendicular Lines

- **Parallel lines** have the same gradient: $m_1 = m_2$
- **Perpendicular lines** have gradients whose product is -1: $m_1 \times m_2 = -1$, so $m_2 = -\dfrac{1}{m_1}$

Example: Find the equation of the line perpendicular to $y = 3x - 1$ passing through $(6, 2)$.

Gradient of given line: $m_1 = 3$. Perpendicular gradient: $m_2 = -\dfrac{1}{3}$.

$y - 2 = -\dfrac{1}{3}(x - 6) \Rightarrow y = -\dfrac{1}{3}x + 4$

Exam Tip: When the question says "in the form $ax + by + c = 0$", multiply through to clear fractions and ensure a, b, c are integers. For example, $y = -\frac{1}{3}x + 4$ becomes $x + 3y - 12 = 0$.

Circles

Equation of a Circle

A circle with centre (a, b) and radius r has equation:

$$(x - a)^2 + (y - b)^2 = r^2$$

The expanded (general) form is:

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

where centre = $(-g, -f)$ and radius = $\sqrt{g^2 + f^2 - c}$.

Example: Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

Complete the square:

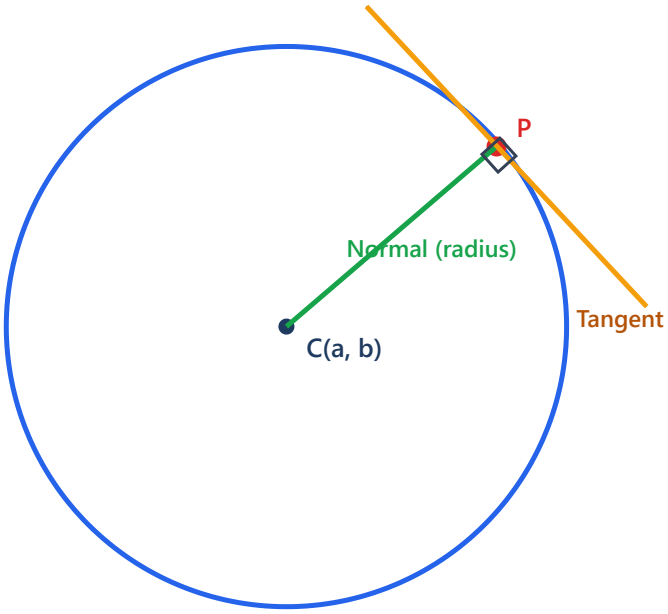
$(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$

$(x - 3)^2 + (y + 2)^2 = 25$

Centre: $(3, -2)$, Radius: 5.

Tangents and Normals to a Circle

- The **tangent** to a circle at a point is perpendicular to the radius at that point.
- The **normal** to a circle at a point passes through the centre (it is the radius).



Tangent is perpendicular to radius at P

Gradient of tangent x Gradient of radius = -1

Example: Find the equation of the tangent to $(x-2)^2+(y-3)^2=25$ at the point $(5,7)$.

Centre: $(2,3)$. Gradient of radius to $(5,7)$:

$$m_{\text{radius}}=\frac{7-3}{5-2}=\frac{4}{3}$$

Gradient of tangent: $m_{\text{tangent}}=-\frac{3}{4}$

Equation: $y-7=-\frac{3}{4}(x-5)$

$4y-28=-3x+15 \Rightarrow 3x+4y-43=0$

Intersection of a Line and Circle

Substitute the linear equation into the circle equation to get a quadratic. The discriminant tells you:

- $\Delta>0$: line intersects circle at 2 points (secant)
- $\Delta=0$: line is tangent to circle (touches at 1 point)
- $\Delta<0$: line does not meet the circle

Example: Show that $y=x+5$ is a tangent to $x^2+y^2=12.5$.

Substitute: $x^2+(x+5)^2=12.5$

$x^2+x^2+10x+25=12.5$

$2x^2+10x+12.5=0 \Rightarrow 4x^2+20x+25=0$

Discriminant: $b^2-4ac=400-400=0$.

Since $\Delta=0$, the line is a tangent to the circle. $\square$

Parametric Equations

Parametric equations define a curve by expressing x and y separately in terms of a third variable (the **parameter**), usually t or θ.

$$x = f(t), \quad y = g(t)$$

Converting Between Parametric and Cartesian Forms

Eliminate the parameter t to get a Cartesian equation relating x and y directly.

Example: A curve is defined by $x = 2t$, $y = t^2 - 1$. Find the Cartesian equation.

From $x = 2t$: $t = \frac{x}{2}$.

Substitute into y: $y = \left(\frac{x}{2}\right)^2 - 1 = \frac{x^2}{4} - 1$.

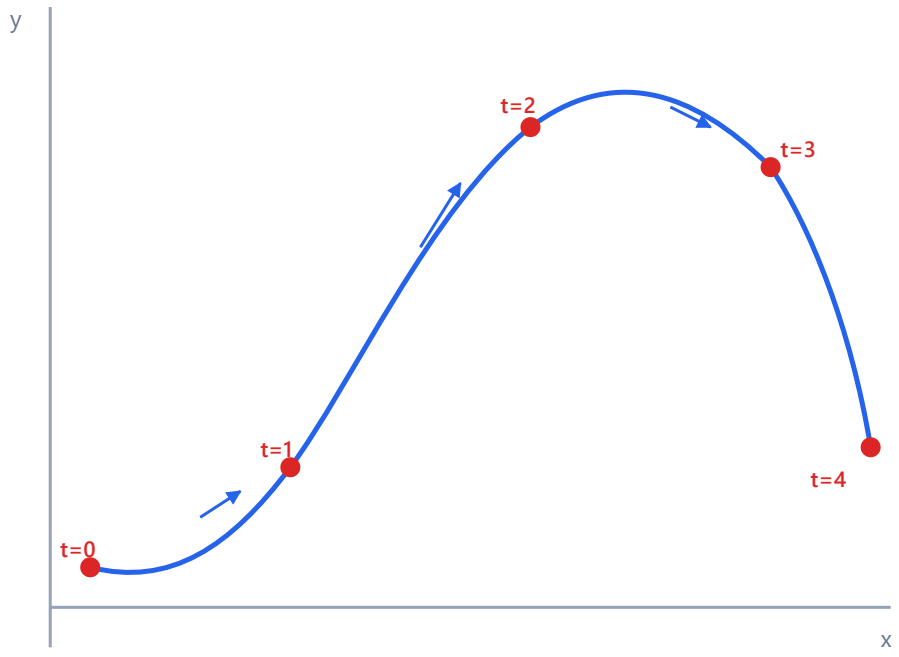
Cartesian equation: $y = \frac{x^2}{4} - 1$ (a parabola).

Example: A curve has parametric equations $x = 3\cos\theta$, $y = 3\sin\theta$. Find the Cartesian equation.

Use the identity $\cos^2\theta + \sin^2\theta = 1$:

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{3}\right)^2 = 1 \Rightarrow x^2 + y^2 = 9$$

This is a circle with centre $(0, 0)$ and radius 3.



Parametric curve: each point corresponds to a value of t

Parametric Differentiation

If $x = f(t)$ and $y = g(t)$, then by the chain rule:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

This gives the gradient of the curve at the point with parameter t.

Example: A curve is defined by $x = t^2$, $y = 2t^3 - t$. Find $\frac{dy}{dx}$ in terms of t.

$\frac{dx}{dt} = 2t,\quad \frac{dy}{dt} = 6t^2 - 1$

$$\frac{dy}{dx} = \frac{6t^2 - 1}{2t}$$

Example: Find the equation of the tangent to the curve $x = 3t$, $y = t^2$ at the point where $t = 2$.

At $t = 2$: $x = 6$, $y = 4$.

$$\frac{dx}{dt} = 3, \quad \frac{dy}{dt} = 2t = 4$$

$$\frac{dy}{dx} = \frac{4}{3}$$

Tangent: $y - 4 = \frac{4}{3}(x - 6) \Rightarrow y = \frac{4}{3}x - 4$

Example: Find the equation of the normal to $x = \sin\theta$, $y = \cos 2\theta$ at $\theta = \frac{\pi}{6}$.

At $\theta = \frac{\pi}{6}$: $x = \sin\frac{\pi}{6} = \frac{1}{2}$, $y = \cos\frac{\pi}{3} = \frac{1}{2}$.

$$\frac{dx}{d\theta} = \cos\theta = \frac{\sqrt{3}}{2}, \quad \frac{dy}{d\theta} = -2\sin 2\theta = -2\sin\frac{\pi}{3} = -\sqrt{3}$$

$$\frac{dy}{dx} = \frac{-\sqrt{3}}{\sqrt{3}/2} = -2$$

Gradient of normal: $\frac{1}{2}$.

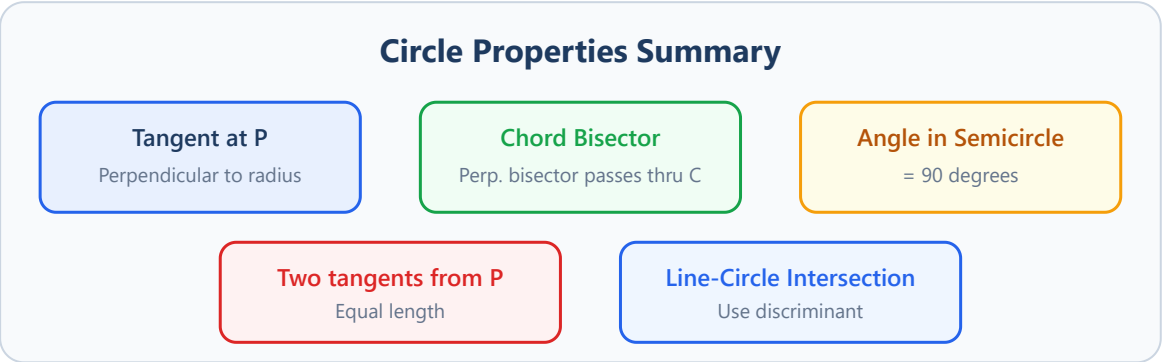
Normal: $y - \frac{1}{2} = \frac{1}{2}(x - \frac{1}{2}) \Rightarrow y = \frac{1}{2}x + \frac{1}{4}$

Exam Tip: When finding tangent or normal equations from parametric forms:

1. Find $\frac{dy}{dx}$ at the given parameter value.

2. Find the (x, y) coordinates by substituting the parameter value.
3. Use the point-gradient form $y - y_1 = m(x - x_1)$.

For the normal, use gradient $-\dfrac{1}{m}$ where m is the tangent gradient.



Circle Geometry Theorems in Coordinate Problems

Several circle geometry theorems are useful tools in A-Level coordinate geometry questions:

- **Angle in a semicircle:** If AB is a diameter and P is any other point on the circle, then $\angle APB = 90^\circ$. This means $\overrightarrow{PA} \cdot \overrightarrow{PB} = 0$ in coordinate terms.
- **Perpendicular bisector of a chord:** The perpendicular bisector of any chord passes through the centre of the circle. Use this to find the centre when given two chords.
- **Tangent from external point:** The two tangents drawn from an external point to a circle are equal in length.

Example: The points $A(1, 6)$ and $B(7, -2)$ are the ends of a diameter. Find the equation of the circle and show that the point $P(7, 6)$ lies on the circle. Verify that $\angle APB = 90^\circ$.

Centre = midpoint of AB = $\left(\dfrac{1+7}{2}, \dfrac{6-2}{2}\right) = (4, 2)$

Radius = $\frac{1}{2}|AB| = \frac{1}{2}\sqrt{(7-1)^2+(-2-6)^2} = \frac{1}{2}\sqrt{36+64} = \frac{10}{2} = 5$

Equation: $(x-4)^2+(y-2)^2 = 25$

P on circle: $(7-4)^2+(6-2)^2 = 9+16 = 25$ ✓

Check $\angle APB = 90^\circ$**:**

$\overrightarrow{PA} = A - P = (1-7, 6-6) = (-6, 0)$

$\overrightarrow{PB} = B - P = (7-7, -2-6) = (0, -8)$

$\overrightarrow{PA} \cdot \overrightarrow{PB} = (-6)(0)+(0)(-8) = 0$ ✓ — confirms $\angle APB = 90^\circ$ (angle in semicircle). □

Example: A circle passes through $A(2,5)$, $B(6,1)$, and $C(0,3)$. Find the centre using perpendicular bisectors.

The centre lies on the perpendicular bisector of AB and of BC.

Perp bisector of AB: Midpoint = (4, 3). Gradient AB = $\frac{1-5}{6-2} = -1$. Perp gradient = 1.

Equation: $y-3 = 1(x-4) \Rightarrow y = x-1$ ...(1)

Perp bisector of BC: Midpoint = (3, 2). Gradient BC = $\frac{3-1}{0-6} = -\frac{1}{3}$. Perp gradient = 3.

Equation: $y-2 = 3(x-3) \Rightarrow y = 3x-7$ ...(2)

Solve: $x-1 = 3x-7 \Rightarrow 6 = 2x \Rightarrow x = 3, y = 2$. **Centre = (3, 2).**

Radius = $|CA| = \sqrt{(2-3)^2+(5-2)^2} = \sqrt{1+9} = \sqrt{10}$.

Equation: $(x-3)^2+(y-2)^2 = 10$.

Exam Tip: If a question gives three points on a circle, use perpendicular bisectors of two chords — their intersection is the centre. If it gives a diameter, the centre is the midpoint and you can use the "angle in semicircle = 90°" property directly.

Exam Strategy Summary

Key points for coordinate geometry questions:

- Always draw a sketch -- even a rough one helps avoid sign errors.
- For circle questions, identify the centre and radius first by completing the square if needed.
- Use the perpendicular gradient relationship $m_1 m_2 = -1$ for tangent/normal problems.
- For parametric differentiation, always use $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$, not the other way round.
- When converting parametric to Cartesian, look for identities you can use (e.g. $\sin^2\theta + \cos^2\theta = 1$).
- Check your answers by substituting back into the original equations.
- Three points on a circle: find the centre by intersecting two perpendicular bisectors of chords.
- Diameter endpoints given: use midpoint for centre, half-distance for radius, and $\angle$ in semicircle = 90°.